

Syllabus Advanced Statistics PSY 4400

Class time: Tuesday and Thursday 2:00 to 3:15
Class location: BSB C231

Lab: Tuesday 3:30 to 4:45 (BSB C105, time primarily reserved for makeup sessions)

Instructor: Keith Sanford, Ph.D.
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Office: BSB A314

Background: I am a clinical psychologist, and I have been developing and teaching this Advanced Statistics course for 25 years. My research pertains to interpersonal relationships, behavioral medicine, and the creation of assessment instruments. Outside of psychology, I enjoy playing music (and I include some music in this course).

Teaching Assistants: Chris Corlett
Email: Chris_Corlett1@baylor.edu

Kun Wang
Email: Kun_Wang1@baylor.edu

Flipped class format

This course will use a type of “flipped” classroom design in that students will complete most instructional material on their own outside of class time, and class time will be used for conducting learning assessments and reviewing material.

How to contact Keith Sanford

Given the structure of this class, it is difficult for me to get to know all the students personally. Therefore, it is wonderful when students contact me outside of class to introduce themselves, to ask questions about the course, or to discuss topics beyond course content. Students are encouraged to stop by my office or contact me to introduce themselves, and to give me an opportunity to ask a few questions to get to know them better. I welcome questions about course content, questions that go beyond course content, or even questions on topics unrelated to psychology just for fun discussion. There are three ways to contact me.

- For ***in-person meetings on campus***, I will be available in my office (BSB A314) from 12:20 to 2:00 every Tuesday. You are welcome to stop by my office during that time, or you can send me an email ahead of time to make sure I am available.
- I will be available for ***Zoom meetings*** at a wide range of times. To request a Zoom meeting, just send me an email listing a few times you are available over the span of a couple days, I can respond by suggesting a time (or requesting more options) and setting up a link for a Zoom meeting.
- I can also respond to questions via ***email***. For example, if you have a quick question about a study guide item or a homework item, you can send me an email with a copy of the item and your question, and I will be happy to respond.

How to contact TAs

The TAs for this course, Chris Corlett and Kun Wang, are graduate students in the Department of Psychology and Neuroscience Ph.D. program. They are an excellent resource for questions about course content and assignments. They will host informal drop-in “office hours” over Zoom at the times posted on Canvas. Students can obtain the links to join these meetings by clicking the Zoom tab in Canvas. The TAs will also be available to answer questions via email or through individual Zoom meetings scheduled through email.

Lab time is primary reserved for makeup sessions

The lab time will be used for two optional instructional sessions covering steps for completing the first two projects (described below), and otherwise, the scheduled lab time for this course will be used only for administering makeup learning assessments. The section below titled, “Missed learning assessments” (starting on page 13), describes specific procedures that students need to follow to schedule a makeup learning assessment if they are absent due to illness or unavoidable circumstance, and it lists specific dates on which makeup assessments will be administered. Makeup learning assessments will only be administered during lab times and only on the scheduled dates. It is essential that all students keep these times reserved on their schedules just case they need to take a makeup.

Course textbook

All the required instructional material for this course will be provided on Canvas, and students do *not* need to purchase a textbook. This course covers the same material as the Howell textbook listed below, which is an excellent statistics textbook and reference. The textbook is listed to clarify the content and difficulty-level of this course. However, this course will differ from the Howell textbook in the way material is organized and presented, and therefore, the textbook would not likely function as a useful study aid for this course.

Howell, D. C. (2013). *Statistical methods for psychology* (8th ed.)

Required materials

Students will need to **bring a charged laptop computer to class each day**. The computer will need Internet access, and students will need to download the program, “Respondus LockDown Browser” on the computer. Students should let me know if they have a life hardship that interferes with their ability to obtain a working laptop computer. For completing projects outside of class, students will need access to a computer and to download the free statistical software programs “R” and “RStudio.”

Course Overview

Course content

This is an advanced course covering statistics for use in psychology and neuroscience research. It is designed for students that have completed introductory statistics, and it will emphasize the application and interpretation of statistics in research settings. Students will learn to use the general linear model as a framework for understanding statistical analysis, and they will learn to calculate both effect sizes (such as: d , r , beta, partial r , R^2 , ΔR^2 , and η^2) and inferential statistics (such as t , and F). This class will focus on statistical analysis in situations where there is a single outcome variable and either single or multiple predictor variables that are either categorical or numerically scaled. Students will gain a solid understanding of important statistical concepts and techniques such as variance, covariance, estimation error, measurement error, sampling error, standardization, regression, multiple regression, focused effects, contrasts, omnibus effects, post-hoc analysis, significance testing, confidence intervals, and analysis of variance.

Course structure

This course will utilize a “flipped” design. Students will complete most instructional material on their own, and they will have access to an extensive series of instructional videos, along with lecture notes, a *Lecture Study*

Guide, homework assignments, and a project manual. There will be a comprehensive, timed learning assessment at the beginning of each class period. Afterwards, class time will be used to play a quiz game that involves review and application of course material. The learning assessments will include questions drawn from four sources: lecture study guides, homework assignments, “Easter egg” questions intended to reward attentiveness to lecture videos, and a list of questions asking for written summaries of results. Students are responsible for watching videos and learning information prior to each learning assessment. Over the course of the semester, students will also complete and submit homework assignments, pre-learning-assessment quizzes, and lab projects. There will be a final exam which will simply be an expanded learning assessment.

Course Components

Learning assessments

There will be an online learning assessment every Tuesday and Thursday at the beginning of each class. The points listed below describe these learning assessments. A later section of the syllabus under the heading titled “When things fail to go as planned” (starting on page 13) describes policies for missed assessments and for laptop emergencies when a student does not have a functioning laptop or when a laptop crashes during a learning assessment.

1. **Laptop computer.** Learning assessments will be completed in class and online. Students will need to bring to class each day a lap-top computer that can be connected to the wireless Internet. The laptop needs to have a functioning battery and students need to make sure they bring the laptop to class with sufficient charge to complete the learning assessment. Although it is OK if students seek ways of utilizing available electrical outlets in the classroom, availability of electrical outlets cannot be guaranteed.
2. **LockDown Browser.** To take a learning assessment, students will need to install “Respondus LockDown Browser” on their laptop computer. (See information below on how to use Respondus LockDown Browser.) It will not be possible to access a learning assessment without this program. At the beginning of each class period, a code number will be displayed at the front of the classroom. To take a learning assessment, students will first start LockDown Browser; then navigate to the learning assessment, then enter the code number to begin the assessment.
3. **Timing of assessments.** A learning assessment will be completed at the beginning of every class day. The first 9 learning assessments will each be a 30-minute timed test that must be completed within the first 35 minutes of class. Starting with learning assessment number 10, all the remaining learning assessments will each be a 35-minute timed test that must be completed within the first 40 minutes of the class period.
4. **Pool of cumulative questions.** The majority of each learning assessment will be comprised of questions drawn from a pool of cumulative questions. Students will have access to the entire pool of cumulative questions that will be used for learning assessments. Specifically, these questions will be drawn from three sources: (a) the *Lecture Study Guide*, (b) homework assignments, and (c) a list of questions asking for written summaries of results. Notably, the items on the learning assessments might not *exactly* match items from these sources. They might have a slightly different appearance, or use slightly different wording, or use different scenarios or different numbers. However, these changes are not intended to be substantial. If you know how to answer all the assigned questions from the study guide, from the homework assignments, and from the list of written summaries (and can answer these questions without using any notes), then you are well prepared to answer most of the questions you will encounter on a learning assessment.

5. ***Increasing difficulty of learning assessments.*** Each learning assessment will emphasize the most recently covered material from the course, but it will also include items from the pool of cumulative questions over all previously covered course content. This means that the pool of possible questions for each learning assessment will grow larger and larger, and the learning assessments will gradually grow more challenging over the semester. The reason for making learning assessments cumulative is because every unit in this course builds on previous units, and students need to remember material covered early in this course to understand material covered later in this course. The amount of cumulative material will vary at different points in the course, but on a typical learning assessment, one-half of the questions might cover recent material, and one-half of the questions might cover cumulative material.
6. ***Easter egg questions.*** In addition to the pool of cumulative questions, the first 17 learning assessments will include questions called “Easter egg” questions. These questions are NOT cumulative and will only pertain to the most recently assigned video. The primary purpose of Easter egg questions is to reward attentiveness to the lecture videos and thereby maximize the likelihood of student success. This is especially important during the first portion of this course when material is easier and students are sometimes tempted to memorize study guide answers without fully understanding the videos. Easter egg questions may test whether students recognize content or examples discussed in the videos, or if students understand the videos well enough to identify the most important points being made. Most Easter egg questions will be multiple choice, and many will ask students to distinguish between something that was part of class lecture versus other options involving things that were never mentioned, or not emphasized, or demonstrably false. For these questions, if more than one response option seems potentially true, the best strategy is to choose an option that you recall being discussed in class lecture (or an option you recall being discussed the most, or one that makes the most important point). There will be 6 Easter egg questions on each of the first nine learning assessments, 4 Easter egg questions on each of the next eight learning assessments (10 through 17), and none on learning assessments after that point. These questions will only be drawn from the main lecture videos and NOT from the “lecture add-on videos” (described below). Easter egg questions will be randomly drawn from an item bank, but unlike the pool of cumulative questions, students will not be given access to the item bank of Easter egg questions.
7. ***Calendar.*** The course calendar (printed at the end of this syllabus and posted on Canvas as a separate document) lists the new content that will be added to each learning assessment on each class day. Every place that the word “Add” appears on the calendar indicates new material that will be added to the pool of items for the learning assessments. Easter egg questions (for the first 17 learning assessments) will be drawn from the video covering the study guide material that was added on that day. Students should be prepared to answer questions covering the new content listed on each day, as well as all the content from the cumulative pool of questions from all previous days.
8. ***Bring a calculator and paper.*** Although the Respondus LockDown Browser program will have a built-in calculator that students can use during learning assessments, students are welcome to bring and use their own calculators (however, students may NOT use their phones as calculators). In addition, students are allowed to bring a single blank sheet of paper and writing utensils to use during learning assessments. Students are only allowed to write on one side of the paper. At the end of each learning assessment, students must put their names on their sheets of paper and turn them in.
9. ***Formulas to memorize.*** There will be a header placed at the beginning of every learning assessment that contains a few formulas you may use as a reference. This learning assessment header is posted on Canvas. Any formulas that you need, and that are not included in this header, are formulas you need to memorize.

10. **Feedback.** After each learning assessment is graded, scores will be posted on Canvas. Copies of graded learning assessments will not be posted on Canvas. As will be described below, student learning is maximized when learning assessment scores are used to provide holistic indicators of overall comprehension, and not item-specific feedback. Item-specific feedback is provided via other mechanisms and not through learning assessments. As will be described below, item-specific feedback is provided through: (a) pre-learning assessment practice quizzes, (b) homework, (c) projects, and (d) in-class activities. Thus, it is not generally expected to be necessary or helpful for students to know which specific items they missed on learning assessments. However, if a student would like detailed feedback regarding her or his performance on a learning assessment, the student is welcome to contact a TA who can look at the student's assessment and give the student feedback on the types of questions that were missed.

Format of items on learning assessments

Learning assessments will include 5 types of items. These include numerical items, multiple choice items, essay items, spelling items, and PsyBU math notation items. Each type is described below.

1. **Numerical answers.** These are questions that are answered by typing a numerical value. When answering these questions on Canvas, it will only be possible to enter numerical values. Unless specified otherwise, **it is acceptable to round these answers to 2 decimal places.** In other words, any answer within .01 of the exact, correct answer will be counted as correct. Note that Canvas does NOT recognize or understand percentage signs. Therefore, if a numerical answer involves a percentage, type the answer **using a decimal and not a percentage sign.** For example, if the answer is 28 percent, DO NOT type "28%," but rather, type ".28" when entering your answer.
2. **Multiple choice.** These are questions that are answered by selecting correct options from a list of possibilities. In some cases, it could also involve matching.
3. **Essay.** These are questions that are answered by typing an answer in a text box. These include questions where you will be asked to write definitions of terms or summaries of results. These questions will need to be graded manually.
4. **Spelling.** Some questions from the *Lecture Study Guide* are fill-in-the-blank questions where the answer needs to be a correctly spelled term. These questions will be automatically graded by Canvas, and so the answers need to be spelled correctly to be counted as correct. The study guide addendum provides a list of correctly spelled terms for each unit.
5. **PsyBU math notation.** Some of the questions from the *Lecture Study Guide* will ask you to write formulas or equations. These will need to be typed using something I call "PsyBU math notation." This is a type of notation that is unique to this class (specifically, it is a simplified version of something called LaTeX notation, which is a widely used system for typing equations using the keys on a standard "QWERTY" keyboard). There is a document posted on Canvas (titled "PsyBU math notation") that describes how to type formulas or equations using PsyBU math notation. Like the spelling questions, the PsyBU notation questions will be graded automatically by Canvas, and so the answers need to be spelled correctly to be counted as correct. The study guide addendum provides a list of correctly spelled formulas and equations for each unit.

Lecture videos

The primary source of instructional material for this course will be in the form of videos that are posted in the "Media Gallery" section of Canvas. There are two types of videos: main lecture videos and lecture add-on videos. There is also a set of lecture notes for each unit and a *Lecture Study Guide* that are posted in the

Modules section of Canvas. In addition to being posted on Canvas, videos are also posted on my Statistics YouTube channel: https://www.youtube.com/channel/UCiseXGmqc8_Ly5DDzcqsIA.

1. **Main lecture videos.** The material for each unit in this class is covered in a set of main lecture videos. Each unit is divided into parts (most have 4 parts labeled Part A through Part D), and each part is covered in a separate video.
2. **Lecture add-on videos.** The lecture add-on videos are intended to be watched after the main lecture videos for a unit, and they have two purposes. First, they provide examples and instruction on how to complete homework questions. Second, they include instructions on how to provide written summaries of results. Students will need to know how to provide these written summaries to complete projects. Thus, students should watch the lecture add-on video for each unit before completing the homework for that unit and before completing the project for that unit. Information from the lecture add-on videos is also important because the learning assessments will include questions drawn from homework assignments and questions that require providing written summaries of results.
3. **Lecture notes.** A set of lecture notes for each unit is posted in the Modules section of Canvas. These notes are different from the *Lecture Study Guide* (described below, which contains questions to answer) and the notes are intended to help students follow along with the lecture videos (including both the main lecture videos and the lecture add-on videos). It is recommended that students have a copy of the lecture notes in front of them when they watch the lecture videos. The notes include a lecture outline, important terms, and copies of figures and statistical output used in the lecture videos.
4. **Statistics music.** The first two units will cover a set of foundational terms that students need to be able to define. For each of these terms, I have written and recorded a song (with a music video) with lyrics that provide a definition of the term. These songs are intended to help students learn the term definitions. When asked to provide a definition on a learning assessment, students have the option of simply writing the lyrics to the appropriate song. These music videos are embedded in the main lecture videos for the first two units, and also posted separately on Canvas, and posted online at www.PsyBU.com.

Lecture Study Guide

The *Lecture Study Guide* includes a set of questions for each of the main lecture videos (but it does not contain questions directly pertaining to the lecture add-on videos). After watching each main lecture video, students should answer the study guide questions for that video, and they should re-watch sections of the video as needed if they are uncertain how to answer any of the questions. Each learning assessment will include questions drawn from the *Lecture Study Guide*. As indicated by the course calendar, a new set of questions from the *Lecture Study Guide* will typically be added to the possible pool of items for each successive learning assessment. The study guide is formatted in a way that students could print the study guide and turn the questions into flash cards to help them study.

Study guide addendum. In addition to the Lecture Study Guide, there is also a study guide addendum posted on Canvas. The addendum provides a list of foundational study guide items that have a high probability of appearing on learning assessments, and it provides lists of correctly spelled terms for the spelling items and the PsyBU notation items that require correct spelling of terms, formulas and equations.

Classroom activity

Whereas the first portion of each class will be used for completing learning assessments, the remaining portion of each class will be used for a classroom activity. The classroom activity will begin when the window for completing the learning assessment closes (which is after 35 minutes of class time for the first 9 learning

assessments, and after 40 minutes of class time thereafter). Students are welcome to leave the room after completing a learning assessment and return for the classroom activity. The classroom activity will include a quiz game in which the class will be divided into teams and students will be called upon to answer questions that involve review and application of course material. While playing the game, students will submit a class activity log on Canvas in which they document for each question: (a) the question number, (b) either their initial answer to the question or a statement that they could not determine an answer, and (c) either a statement their initial answer was correct or the correct answer.

Pre-learning-assessment practice quizzes

Prior to each learning assessment (except for the very first one) students will take an at-home pre-learning-assessment practice quiz. This practice quiz will contain a maximum of 15 questions that are all: (a) drawn from the study guide, (b) identical copies of questions that appear on learning assessments, and (c) questions that involve typing formulas or equations using PsyBU notation (and occasionally, questions that involve describing distributions). The purpose of these practice quizzes is to provide item-specific feedback to students regarding questions they are missing, and to encourage students to engage in regular study and rehearsal of course material. These practice quizzes will focus on formulas and equations because this is the material that is most frequently missed on learning assessments. Each practice quiz must be completed during a one-week window that ends at 11:59 PM (one minute before midnight) the day before a learning assessment. To gain maximum benefit in preparation for actual learning assessments, students are encouraged (but not required) to complete practice quizzes without using their notes. Each practice quiz may be taken multiple times, and when taken multiple times, the highest score is used for grading. (But, do not repeat a practice quiz after the due date because this will confuse the Canvas system and it will try to impose a late penalty).

Homework

At the end of each unit, students will complete a homework assignment. Due dates for each homework assignment are listed in the course calendar. All assignments are submitted on Canvas and due by 11:59 PM (one minute before midnight) on their listed due dates. Assignments are graded automatically. On Canvas, students will be able to identify any missed items immediately after submission, and they will be able to obtain correct answers for those items starting one day after the due date. The lecture add-on videos provide examples of how to complete homework items, and students should watch the lecture add-on video for each unit before completing the homework assignment. Materials for completing each homework assignment are located on Canvas. Specifically, there is a link to download a copy of the homework questions in the Modules section of Canvas, and there is a submission portal to submit answers in the Assignments section of Canvas. Most students find it is easiest to download the questions, work out all the answers on paper, and then log onto Canvas and submit answers. The submission portal is set up so that all items are answered using either a multiple-choice format or a numerical answer format. As with the learning assessments, unless specified otherwise, it is acceptable (but not required) to round numerical answers to **2 decimal places**, and if a numerical answer involves a percentage, it should be entered using a decimal and **NOT with a percentage sign**.

Projects

Students will complete a series of four projects which will provide them with experience in multiple aspects of conducting research. This includes experience with managing data files, conducting statistical analyses, interpreting results, and writing up results. Due dates for each project are listed in the course calendar and all projects are due by 11:59 PM (one minute before midnight) on their listed due dates. To complete these projects, students will need to download and use the free statistical software programs “R” and “RStudio.” R is probably the most powerful, flexible, and highly regarded statistical software program currently available. However, unlike other menu-driven statistical software programs that are easy to use, R is more complex, and requires entering commands using computer programming language. This means that knowing how to use R is a marketable skill. Other advantages of R include the fact that it is free, and compared to other programs, the process of learning R leads to more in-depth understanding of statistical analysis. Detailed instructions on how

to download R and how to complete each project are provided in a project manual posted in the Modules section on Canvas. In addition, each project requires students to provide written summaries of results, and instructions on how to write these summaries are included in the lecture add-on video for each unit. It is advisable to complete projects early and not wait until the day a project is due before completing it. In running the data analysis for projects, it is not uncommon for students to make computer coding mistakes that produce unexpected error messages that can be time consuming to resolve. If needed, a TA can provide help in these situations, but it may not be possible to provide help on the same day a project is due.

There are two project lab sessions listed on the course calendar. These are optional sessions for students that would like to initiate work on their first couple projects in a group setting with in-person instruction. These sessions are available to provide students with additional guidance in learning how to use the R program.

Final exam

The final exam for this course will simply be an extended learning assessment. Instead of a learning assessment with a 35-minute time limit, the final exam will be a learning assessment with a 2-hour time limit, and it will be about three times longer than a typical learning assessment.

Examples Illustrating How to Take This Course

To provide clarification on the process of completing this course, this section provides examples illustrating what you need to do. Look at the course calendar. Any place the word “due” appears on the calendar indicates that an assignment is due on that day and the assignment must be submitted on Canvas before the day is over (by 11:59 PM, one minute before midnight). Any place the word “Add” appears on the calendar indicates that new material is being added to the pool of cumulative items for the learning assessments. Make sure you are prepared to answer questions over this new material and all preceding material. *For example, . . .*

1. For August 28, the calendar says: Add Study Guide 0A-1A questions.

To prepare for the learning assessment on this day, first study *Lecture Study Guide* questions 0A which pertain to information printed in this syllabus. Then, you should print a copy of the lecture notes of Unit 1, watch the main lecture video for Part A of Unit 1, and study the *Lecture Study Guide* questions for section 1A. If the *Lecture Study Guide* has any questions asking for formulas or equations, make sure you can answer those questions using PsyBU math notation. Also, use the study guide addendum to make sure you have the correct spelling for items that require correct spelling or PsyBU notation, and to identify items that may be especially important. The learning assessment on this day will be drawn from questions from sections 0A and 1A of the *Lecture Study Guide*, and it will include six Easter egg questions from the video for unit 1A.

2. For September 1, the calendar says: Begin practice quizzes.

This means you need to complete a pre-learning-assessment practice quiz before 11:59 PM on this day. From this point on, a pre-learning-assessment practice quiz needs to be completed by 11:59 PM one day prior to every learning assessment. To complete each practice quiz, you should be prepared to answer all the formula and equation items from the study guide that have been covered up to that point in time.

3. For September 2, the calendar says: Add Study Guide 1B questions.

To prepare for the learning assessment on this day, you should watch the main lecture video and study the *Lecture Study Guide* questions for Part B of Unit 1. The learning assessment will include questions from units 1B, 1A, and 0A of the study guide, and it will include six Easter egg questions from unit 1B.

4. For September 8, the calendar says: Homework 1 due.
This means you need to watch the lecture add-on video for Unit 1, complete the homework for Unit 1, and submit the homework on Canvas before 11:59 PM on this day.
5. For September 9, the calendar says: Add Study Guide 2A questions, Add Homework 1 questions
This means that you should watch the main lecture video and study the *Lecture Study Guide* questions for Part A of Unit 2, **and** you should be prepared to answer any question taken from Homework 1. The learning assessment will contain questions from section 2A of the study guide, six Easter egg questions from unit 2A, questions from Homework 1, and cumulative questions from all the previous material that has been added to the pool of items. There will also be an optional lab session on this day where students can begin work on Project 1 in a group format with in-person instruction.
6. For September 10, the calendar says: Suggested try Project 1
Nothing is actually due on this day, but it is **highly recommended** that you attempt to complete Project 1 by this day. Students may complete the project entirely on their own or attend the project lab session listed on the calendar for additional instruction. To complete the project, follow instructions in the lab project manual posted on Canvas. The project involves installing and using the R computer software program, and when students first attempt to use this program, they sometimes encounter errors. If you attempt to complete the project on this day and encounter errors that you cannot figure out, you will have time to obtain help before the project is due.
7. For September 15, the calendar says: Project 1 due
This means that Project 1 needs to be submitted on Canvas before 11:59 PM.
8. For September 18, the calendar says: Add Study Guide 2D questions, Add Unit 1 written summaries
This means that you should watch the video and study the *Lecture Study Guide* questions for Part D of Unit 2, **and** you should be prepared to provide a written a summary of results as described in the Lecture add-on video for Unit 1 (these summaries will be the same as summaries you wrote as part of Project 1). You should also be prepared to answer six Easter egg questions from unit 2D and cumulative questions from all the previous material that has been added to the pool of items.
9. And so forth.

Important Things to Know About This Course

Five tactics for doing well in this course.

1. ***Engage in regular study and review of course material.*** After you watch each video, answer all the corresponding study guide items. Then, turn the items into flash cards that you can review and rehearse. After you complete each homework, go back, and review all the homework items, and make sure you can complete all the items without using your notes. After you complete each project, go back, and review the templates for written summaries. Make sure you have these templates memorized and that you can produce a written summary without using your notes. On a regular basis, practice doing study guide items, homework items, and written summaries. Make sure that you keep up with the material. Given the cumulative nature of material in this course, it is quite difficult to catch up if you get behind. Students that fall behind early in the course often find themselves overwhelmed and confused by the end of the course. If you engage in regular study and review, the course will be much easier, and you will learn a substantial amount of information about statistics.

2. ***Find answers to your questions.*** If at any point you are not sure how to answer a study guide item, or a homework item, or how to do a written summary, go back, and review the video. If you are still uncertain about an item, ask about it during class, or in an individual meeting with a TA or with me. On the one hand, you will learn material best if you first attempt to answer questions on your own. On the other hand, there are likely to be times when you will still have questions even after reviewing the videos. If you have made a sincere effort to determine the correct answer on your own, I am always happy to answer questions to make sure you have the correct answers.
3. ***Develop your “judgment of learning.”*** Judgment of learning is the metacognitive skill of being able to determine for yourself how well you know and understand the material being covered in a course. Several components of this course are designed to help you develop judgment of learning. You will receive feedback from learning assessments, pre-learning-assessment quizzes, homework assignments, and projects. You will also have access to the pool of cumulative questions, which means you can test yourself to see how well you know the information and identify any items giving you trouble. Taken together, these resources are sufficient for students to develop judgment of learning and to identify specific areas of strength and weakness. However, this process requires effort. Students need both to make use of the feedback they receive and to test themselves. You will maximize your success in this course by using the available resources and devoting effort to develop your own judgment of learning.
4. ***Focus on the concepts.*** Focus your efforts on understanding the big-picture concepts covered in this class and avoid excessive reliance on rote memorization. When you answer questions on the study guide and on homework assignments, it is best if you can answer questions based on your understanding of the concepts behind the questions and on your understanding for why particular answers are correct. Given that you will have access to the entire pool of cumulative questions, you may be tempted to try a strategy in which you attempt to memorize answers to questions without fully comprehending the information. This strategy might work in the short term but would likely prove disastrous in the long term. This is because each unit builds on previous units, all learning assessments are cumulative, and the range of content covered on each successive learning assessment gets larger and larger. If you attempt to learn this material through rote memorization, it will soon become overwhelming. However, if you understand the concepts, then you can use your understanding of the concepts to answer all the questions.
5. ***Make full use of the videos.*** One key to making sure you understand the concepts is to use the videos. Watch the videos with a goal of understanding concepts and not merely answering study guide items. On the one hand, the study guide is intended to facilitate learning of concepts. On the other hand, it cannot guarantee you have learned the concepts. At times, it may be tempting to skip over parts of videos if you have identified correct answers to study guide items. However, you will create substantial problems for yourself if you skip parts of a video without fully understanding the concepts being covered. Often, the concept covered in one video is something that will come up over and over again throughout the course. Thus, if you fail to understand a concept in one video, it will make future units extremely difficult, and this will be true regardless of whether you can answer study guide questions. If you watch the videos with a goal of understanding the concepts, it will make future units much easier.

Course components designed to facilitate student learning.

Several components of this course are designed to facilitate student learning and help students follow the tactics for doing well that are listed above.

1. ***Frequent learning assessments.*** The primary reason for having a learning assessment during each class is to encourage students to engage in regular study and review of course material.

2. ***Highly edited videos.*** The videos for this course are not merely screen captures of live course lectures. Each video has been planned, rehearsed, recorded multiple times, and then edited to produce a final video lecture that is as concise and clear as possible. In general, each hour of video took over 10 hours to create. The videos are intended to provide a comprehensive coverage of material for this course. Because I created the videos and all the material for this course myself, this means that students do not need to spend money to purchase textbooks and study materials from other authors or other companies. The videos are designed to provide students with a solid understanding of statistical concepts, and to provide a resource so students can find answers to their questions.
3. ***Access to the pool of cumulative questions.*** As described above, the lecture study guide, the homework assignments, and the list of written summaries provides students with the entire pool of cumulative questions that will be used for learning assessments. This means that students know exactly what the learning objectives are for the course and they know how they need to study for the learning assessments. Moreover, this gives students the tools they need to test themselves and to develop judgment of learning.
4. ***Components to encourage effective study habits.*** The Easter egg questions are intended to reward students for making full use of the lecture videos. The pre-learning-assessment practice quizzes are intended to help students engage in regular study and rehearsal of material, and to provide item-specific feedback on the types of items that are most frequently missed and most difficult to remember.

Effective use of feedback requires work

Students in this course are given access to an extensive set of resources they can use to test their knowledge. Students receive item-level feedback from 26 pre-learning-assessment quizzes, 6 homework assignments, and 4 projects. They receive global-level total scores from 27 learning assessments. They are given access to the entire pool of cumulative questions that may appear on learning assessments, and by following the tactics for doing well in this course (described above), they should be able to determine all the correct answers for those questions.

Although students receive an extensive amount of feedback and guidance in this course, they may not always experience this feedback as positive. There are two key reasons for this. First, feedback can heighten a student's awareness of things the student does not know, and this can create feelings of frustration and heighten anxiety about having correct answers. Second, it requires work to make effective use of feedback, and if a student is not working to use the feedback, the feedback will feel demoralizing, and it will not be perceived as feedback. A paradoxical trend I have observed in my teaching is that the more I provide students with study aids and feedback, the more I *increase* the risk that some students will experience anxiety about knowing correct answers and end up perceiving there to be a *lack* of guidance and feedback. It is illustrative to note that, across many years of teaching, I have sometimes taught courses in which I gave students no study guide whatsoever, and I simply told them that anything from lectures and assigned readings could appear on tests. In these courses, I have rarely ever had students complain about not being given correct answers to test questions or about not receiving enough feedback. In contrast, the most common complaints I am likely to receive from students in this Advanced Statistics course will be complaints about not receiving enough feedback, not receiving the answer key for the pool of test items, and not receiving item-by-item feedback on learning assessments. These types of complaints reflect genuine forms of frustration that some students are likely to experience with this class. Hopefully, I can reduce the likelihood of this frustration by calling attention to this risk up front.

The forms of feedback and guidance employed in this course are specifically calibrated to provide an ideal level of challenge to maximize student learning. Specifically, student learning is maximized when there is a moderate level of challenge to the process of obtaining correct answers to questions. If it is too difficult to obtain correct answers, students may become frustrated and give up. However, if it is too easy, it will sabotage student

learning. In this particular course, if it is too easy for students to obtain correct answers, it destroys student motivation to: (a) develop one's own judgment of learning, (b) focus on concepts, and (c) make full use of the videos. If all the correct answers were readily available, it would create a strong temptation for students simply to memorize the correct answers to the questions without watching the videos, without comprehending the concepts, and without investing effort to develop one's own judgment of learning. In addition, there is a human tendency to overestimate one's own ability after being shown correct answers, and this can lead to reductions in study effort. The more effort invested in seeking to obtain correct answers, the better the material is learned and retained. In sum, if it is too easy to obtain correct answers, it can destroy student motivation for engaging in several key tactics that are essential for doing well in this course. This would lead to substantial numbers of students becoming overwhelmed and confused as material becomes increasingly more difficult over the course of the semester. Thus, students need to experience some degree of challenge to obtaining correct answers. This is especially important in this Advanced Statistics course because it includes an extensive set of resources that provide guidance and feedback.

To ensure that students experience sufficient challenge to the process of obtaining correct answers, there are two areas where students are, intentionally, NOT given automatic access to a set of correct answers.

1. ***Students will NOT be given a complete answer key for the lecture study guide.*** Although there are many ways that students can obtain correct answers and test their knowledge (as described in sections above), an answer key for the study guide will not be posted.
2. ***Learning assessments not released.*** After each learning assessment is completed, scores will be posted on Canvas, but copies of graded learning assessments will NOT be posted, and students will NOT be given automatic feedback on which specific items they missed. Although it is possible for students to obtain item-level feedback for a learning assessment (as described in sections above), the learning assessment scores are intended to provide a holistic indicator of student performance in class. They are not intended to provide item-by-item feedback about exactly which questions a student missed and which ones were answered correctly. Item-level feedback should be gained through the 26 pre-learning-assessment quizzes, through the 6 homework assignments, through the 4 projects, by asking questions in class, by meeting with me or a TA, and through testing yourself to develop your own judgment of learning. If you receive a score on a learning assessment that is lower than desired, it should be treated as a holistic indicator of a potential problem (which can be addressed by increasing use of the tactics for doing well in this course described above). If you are uncertain why you received a low score, it could indicate a need to increase your own judgement of learning. This is best accomplished by using the resources that provide item-level feedback and by testing yourself using the pool of cumulative items. This approach takes more effort than if a student is simply told which specific items were missed on each learning assessment. For some students, this may create anxiety or lead to a perception of not receiving guidance and feedback. However, the rationale for this policy is that students will learn the material better when students invest effort in utilizing the provided resources to develop their own judgment of learning.

Reliability of learning assessments

Students should also be prepared for the fact that random factors can cause fluctuations in scores on learning assessments. This is something that is true for all tests in any class, and it is related to the issue of test reliability (which is something we will cover in this class). Random score fluctuations can be the result of a student making a careless error, or typo, or misunderstanding a question – things that happen to everyone from time to time. Other times, a student might happen to make a lucky guess. Also, a student might happen to get a randomly selected Easter egg question that is easier or more difficult than other items from the bank. It would not be uncommon for a student to get a score that is 8 points higher or 8 points lower simply due to these types of random factors. Fortunately, this issue is effectively addressed by the fact that students complete many

learning assessments. Averaging across multiple learning assessments scores will produce a much more reliable estimate of your performance than looking at a single score. This is another reason why it is important to treat your learning assessment scores holistically and why it is important to use other resources to develop your own judgment of learning.

When Things Fail to Go as Planned

document@PsyBU.com

An email account, with the address “document@PsyBU.com,” is set up to hold documentation of situations where extenuating circumstances interfere with a student’s class participation (for example, if a student cannot attend class due to an illness). The sections below will describe specific situations where you should send an email to this address. When the TAs or I need to review student documentation pertaining to extenuating circumstances, this is the place we will look for it. If needed documentation is not sent to this account, there is a risk it will be overlooked. Note, however, that I do not check this email account on a regular basis, and I typically do not send personal replies to messages sent to this account. If you need to talk with me about something, please send a message directly to my regular email address: Keith_Sanford@baylor.edu.

Laptop emergencies

If a student comes to class without a functioning laptop (due to computer malfunction, accident, or other emergency), or if a student’s computer crashes during a learning assessment, the student needs to take necessary steps to ensure that the problem is fixed as quickly as possible so that it does not continue in the future. In this situation, a student will be allowed to take a paper-and-pencil version of the assessment during class; however, the student will need to submit a “documenting email” as described below.

Documenting email. Before the end of the day on which a paper-and-pencil assessment was completed, the student must send an email to document@PsyBU.com that both: (a) explains the situation that resulted in the need to take a paper-and-pencil version of the learning assessment, and (b) explains concrete steps the student is taking to fix the problem as fast as possible to ensure it does not continue in the future. If a student needs to take a paper-and-pencil assessment on more than one occasion, a separate documenting email needs to be sent on every occasion, and each email must be different. In other words, it must describe a different situation and a different solution, or if the same problem reoccurs across multiple days, the email must explain why the problem is reoccurring and how the student is taking **new** steps to address the problem. If a documenting email is not sent by the end of the day, a 5-point penalty will be applied to the T-score grade for the learning assessment.

Missed learning assessments

This section describes what to do if you need to miss a learning assessment due to an unavoidable situation. Unavoidable situations that may cause students to miss class include an illness, an emergency, a grad-school interview, or participation in a university sponsored activity. When students meet all the criteria and take all the steps listed below, they will be allowed to make up missed learning assessments during scheduled makeup sessions. Otherwise, missed learning assessments will be assigned a T-score of zero. If you need to miss a learning assessment due to an unavoidable situation, follow the steps and policies described below.

1. **Provide notification before the start of class.** Before the start of a class that you need to miss, send a notification email to document@PsyBU.com that states the reason for the absence. To ensure that this email gets recorded and counted, it needs to: (a) have the word “absence” in the subject line, (b) state the date of the missed learning assessment, and (c) state the learning assessment number – this is the number that is listed after each date on the course calendar.
2. **Type answers to study guide questions.** To ensure that you are keeping up with course material, type your answers for all the questions in the section of the study guide that is listed on the calendar as being

added to the learning assessment that you missed. This is in lieu of participation in the classroom activity, and you will need to have your typed answers to submit a “Request for Makeup” form.

3. ***Submit a “Request for Makeup” form before midnight on the day after you return to class.*** A copy of the “Request for Makeup” form can be downloaded from the course documents section of Canvas. Complete the form and submit it to document@PsyBU.com with the name of the requested makeup session in the subject line. Note that you will need to paste your written study guide answers into this document. It is due by midnight on the day after you return to class. If a single unavoidable situation causes you to miss sequence of back-to-back learning assessments in a row, then a separate form needs to be submitted for each learning assessment, and **all** the forms are due on the day after you return to class.
4. ***Collect and save documenting evidence.*** You do NOT need to provide documenting evidence to submit a “Request for Makeup” form; however, a penalty is applied if you submit **more than 4** during the semester, and you WILL need documenting evidence if you want to submit a waiver request to have this penalty removed (waiver requests are described on page 15). Types of documenting evidence may include a note from a healthcare professional, notice of requirement to quarantine, letter from university chaplain or social worker, official notice of participation in a university sponsored activity, copy of a funeral service, or a copy of an invitation to a graduate school interview. Save this information so that you have it in case an unavoidable circumstance causes you to miss more than 4 classes and you need to submit a waiver request.
5. ***Schedule of learning assessment makeup sessions.*** Makeup learning assessments will only be administered during lab times and only on the scheduled dates listed below. Only one learning assessment can be completed at each makeup session.

Session A: September 16	Session F: November 11
Session B: October 7	Session G: November 18
Session C: October 21	Session H: December 2
Session D: October 28	Session I: December 9 during class time
Session E: November 4	Session J: December 9 during lab time

6. ***Late makeup sessions.*** A missed learning assessment needs to be completed at the earliest available makeup session, otherwise it will be considered late. Specifically, the makeup learning assessment must be completed during the next available makeup session after the due date for submitting the “Request for Makeup” form (which is at midnight on the day after you return to class). If a student has more than one learning assessment to make up, the assessments must be completed during successive makeup sessions.
7. ***Limit to the possible number of makeup learning assessments.*** Students can only make up learning assessments that fit within the time slots listed above in the “schedule of learning assessment makeup sessions.” If a student misses more learning assessments than can be made up in the available time slots, then any missed assessments exceeding the available time slots cannot be made up. If these missed assessments are due to a documented unavoidable life circumstance, the student may discuss with me the option of taking an incomplete grade and establishing a plan to make up assessments after the end of the semester.
8. ***Possible penalties applied to makeup learning assessments.*** Makeup learning assessments are subject to the following penalties. These penalties are point reductions applied to the T-score learning

assessment grade (Canvas will label all these penalties as “late assignment” penalties regardless of the cause).

- 10-point penalty to each requested makeup learning assessment that exceeds a total limit of 4 makeup requests for the semester. (Students may request a waiver to this penalty by following procedures described below).
- 10-point penalty if the answers to study guide questions submitted on the “Request for Makeup” form are not adequate (for example, if answers are missing, if they lack question numbers, or if they contain too many mistakes).
- 5-point penalty if the scheduling request form was submitted late, but no more than one week late.
- 20-point penalty if the scheduling request is submitted more than one week late.
- 5-point penalty if the student did not send a notification email prior to the start of class.
- 5-point penalty if the student completes a late makeup session.
- 5-point penalty if the student makes minor, careless errors on the “Request for Makeup” form. (Substantial errors could result in the request being denied altogether.)

Late assignments

All homework assignments, pre-learning-assessment practice quizzes, and projects are due at 11:59 PM (one minute before midnight) on the due dates listed in the course calendar. Canvas is set up to apply a late assignment policy to all homework assignments, practice quizzes, and projects turned in after 11:59 PM. Specifically, Canvas will deduct 5% from assignments less than an hour late, and it will deduct an additional 5% for each additional hour an assignment is late, but it will not deduct more than 90%. This means that an assignment can always be submitted any time after it is due, but if it is more than 17 hours late, Canvas will score it as being worth only 10% of its original value. If a situation arises where you are not able to turn in an assignment on time and you have a reason to believe that you should be excused from the late assignment penalty, then submit the assignment as soon as it is feasible to do so and submit a request to waive the penalty (following the instructions below).

Waiver Requests

If you incur a penalty of any kind (on a learning assessment, homework, practice quiz, or project) and you believe you should be excused from the penalty, you may submit a waiver request. Below are examples of situations where students may submit waiver requests.

- A student submitted more than 4 “Request for Makeup” forms for missed learning assessments, and the student has documenting evidence demonstrating that the absences were unavoidable.
- In a one-time event, a student needed to miss a learning assessment and submitted a notification email that was a little late because the student was too sick to submit the email prior to the start of class.
- In a one-time event, a student submitted a homework assignment a little late due to a family emergency.

Follow the procedures listed below to submit a waiver request.

1. All waiver requests should be sent to document@PsyBU.com with the words “waiver request” in the subject line.
2. If the waiver request pertains to material being submitted late, the late material should be submitted as soon as it is feasible to do so. If it pertains to the late submission of an assignment, the assignment should be submitted as soon as possible. If it pertains to a late submission of a notification for an absence, the notification should be submitted as soon as possible. The sooner the late material is submitted, the greater the likelihood that the waiver will be granted.

3. Include any relevant documenting evidence in your waiver request (such as a note from a healthcare professional, university chaplain, or social worker, or a notice of a university sponsored activity, funeral, or graduate school interview). Documenting evidence is required if a student is requesting a waiver from the penalty for submitting more than 4 “Request for Makeup” forms. For all other types of waiver requests, the need for documenting evidence depends on the nature of the situation.
4. In most cases, waiver requests will be saved until the end of the semester and decisions about granting waiver requests will be made at the end of the semester. This allows decisions to be made based on a complete understanding of the student’s situation across the entire semester, including the extent to which the waiver request involves a one-time event or a repeated problem. Also, by making these decisions all at once, it is possible to make decisions that are consistent and fair for all students.
5. Submission of a waiver request is not a substitution for direct communication with me about an important issue that needs to be discussed in person. Emails submitted to document@PsyBU.com are automatically placed in a folder that is used for grading decisions, and this is NOT a folder I review on a regular basis. If the waiver request involves an important issue to discuss with me in person, then contact me directly in addition to submitting a waiver request.

Contingency Plans

It is possible that a learning assessment will be cancelled, for example if the university closes campus due to inclement weather or if the Baylor server system is down at the time of a scheduled assessment. In these situations, the cancelled assessment will be rescheduled to occur on the next class day along with the learning assessment that was originally scheduled for that day. This means that, if a learning assessment is cancelled, on the **next** class day students will complete two learning assessments (and there will not be any other classroom activities on that day). In this way, a single cancelled assessment will not otherwise disrupt or alter subsequent dates on the course calendar.

Units

Unit 1	Introduction, General Linear Model, and Variance	Unit 4	Simple regression
Unit 2	Effect Size and Types of Error Distributions	Unit 5	Multiple regression
Unit 3	Simple Contrasts	Unit 6	Multiple contrasts

Grading

All grades will be posted on Canvas. This includes grades for learning assessments, homework assignments, and projects.

Although this course is listed as including a lecture component (PSY 4400-01) and a lab component (PSY 4400-A), this distinction will not be utilized in assigning grades. At the end of the semester, each student will be assigned a single final grade that applies both to the lecture component and to the lab component of this course.

Grading scale. This course does NOT use the percentage grading system that may be familiar to most students. Instead, grades will be assigned using the following point system.

Grade	Points
A	58
B+	54
B	50
C+	45
C	33
D	24
F	< 24

The above chart indicates the MINIMUM value needed to obtain each letter grade. Scores will NOT be rounded up at the end of the semester. Thus, a final score of 57.96 would receive a grade of “B+” and not an “A.”

Learning Assessment scores. Scores for all learning assessments (and the final exam) will be given as T-scores, with a mean of 50 and a standard deviation of 10. These can be transformed into letter grades using the grading scale above.

Note that Canvas may *incorrectly* state that learning assessment T-scores are “Out of 100 points.” This is because the Canvas program has difficulty handling T-scores and needs to assign an arbitrary, meaningless point value. Learning assessment grades are given as T-scores and they are NOT “Out of 100 points.” If any penalties are applied to a makeup learning assessment or to a paper-and-pencil learning assessment (as described above), Canvas will *incorrectly* label the penalty as a “late” penalty regardless of the true reason for the penalty.

Details regarding the calculation of T-scores include the following. T-scores will be based on means and standard deviations that are calculated after removing any student earning 25% or less of the possible raw-score points. If some learning assessments, especially near the beginning of the semester, are relatively easy, it is possible that many students will earn perfect scores, and this will have the effect of lowering the ceiling for the T-scores. To address this issue, the ceiling for T-scores will not be allowed to fall below 61. If a T-score conversion results in a situation where a perfect score is something less than 61, then a constant will be added to all student’s T-scores so that a perfect score is equal to 61.

Homework, projects, and pre-learning-assessment practice quizzes. Students are expected to achieve a high degree of accuracy when completing homework assignments, projects, and pre-learning-assessment practice quizzes. These scores will be computed using a procedure different from the procedure for calculating learning assessment scores. Specifically, these scores will be calculated as the percent correct out of 61 possible points, and these scores can be transformed into letter grades using the grading scale above. Scores on projects will be converted to this scale prior to posting. Scores on homework assignments and practices quizzes will be posted on Canvas as raw scores and then converted to this scale at the end of the semester.

Classroom activity score. Students will receive four classroom activity scores, with each score pertaining to a block of approximately seven classes. Each score will be based on the percentage of classroom activities attended, out of 61 possible points, minus reductions if there are deficiencies in the quality of the submitted

activity logs. When calculating the percentage of classroom activities attended, students will automatically be excused from absences on days for which a “Request for Makeup” form was submitted. Thus, this score will be determined by first removing excused absences; then using the remaining classes to calculate the percent of class activities that the student attended. The quality of submitted class activity logs will be based on a random audit of class logs. At least one log will be audited for each student, and more will be audited if any issues are observed. A 5-point penalty will be applied to the entire block class activity score for each problem or issue that is uncovered during the audit (for example, a quiz item lacking sufficient documentation or lacking a correct answer). If the audit reveals one or more submitted logs with *substantial* issues (for example, a log lacking correct documentation for more than 50% of the quiz questions), the entire block will be assigned a score of zero. Otherwise, the score will be calculated as 61, times the percent of classroom activities attended, minus any penalties from the audit.

Final grades. Final scores will be calculated as the average of 46 grading units and transformed into letter grades using the grading scale above. The 46 grading units include the following.

1. There are 27 learning assessments, and each will count as 1 grading unit (27 units all together).
2. The final exam will count as 4 grading units.
3. There are 4 projects, and each project will count as 1.5 grading units (6 units all together).
4. A single total homework score will be calculated, and it will count as 4 grading units.
5. A single total score will be calculated for the pre-learning-assessment practice quizzes, and it will count as 3 grading units.
6. There are 4 classroom activity scores, and each will count as one-half unit (2 units all together).

Thus, to compute final scores, the following 6 components will be calculated:

Component 1 = (Sum of 27 learning assessments) (1)

Component 2 = (Score on final exam) (4)

Component 3 = (Average of 4 projects) (6)

Component 4 = (Total homework score) (4)

Component 5 = (Total pre-learning-assessment quiz score) (3)

Component 6 = (Average of 4 classroom activity scores) (2)

Then, these 6 components will be added together, and the sum will be divided by 46.

Estimating grades during the semester

At any point during a semester, you can use the formula below to estimate your current grade in the class

$$\text{score} = \frac{(\overline{LA})(31) + (\overline{PR})(6) + (HW)(4) + (PreLA)(3) + (\overline{CA})(2)}{46}$$

Where:

$\overline{LA}$ = Your average learning assessment score across all learning assessments completed to date.

$\overline{PR}$ = Your average project score across all projects completed to date.

$$HW = 61 \left(\frac{\text{total homework points earned to date}}{\text{total possible homework points to date}} \right)$$

$$PreLA = 61 \left(\frac{\text{total pre learning assessment quiz points earned to date}}{\text{total possible pre learning assessment quiz points to date}} \right)$$

$\overline{CA}$ = Your average classroom activity score across all scores posted to date.

Important University Policies

Academic Integrity

It is considered cheating for a student to take or submit a learning assessment from a location outside the classroom. It is also cheating to complete or submit a log for the in-class activity from a location outside the classroom. Attendance will be taken to ensure that students submitting learning assessments are present in the classroom. If you are present in the classroom, but not sitting in your assigned seat, it is your responsibility to make sure a TA or the instructor is aware of this fact. Otherwise, you be counted as absent, and any submitted in-class work will be viewed as evidence of cheating. In addition, any work submitted under your name is expected to be your own, neither composed by anyone else as a whole or in part, nor handed over to another person for complete or partial revision. Instances of cheating, plagiarism, or any other act of academic dishonesty, will be reported to the Honor Council and may result in failure of the course. As a Baylor student, you are expected to be intimately familiar with the Honor Code at: <https://www.baylor.edu/honorcode/>.

Students Needing Accommodations

Any student who needs academic accommodations related to a documented disability needs to communicate with me directly. I do not implement accommodations based on automated emails I receive from OALA unless a student directly communicates with me. A commonly requested accommodation for this class involves taking learning assessments at the OALA testing center. To implement this accommodation, students need to start each class early at the OALA testing center, allowing enough time to complete the learning assessment and then walk to class for the classroom activity. Due to feasibility issues, once learning assessments have been moved to the OALA testing center they cannot be moved back to the classroom. Of the students entitled to this accommodation, most find it unnecessary, a few find it helpful later in the semester, and it is very rare for students to need this accommodation at the beginning of the semester. If I am given a week of advance notice, I can implement this accommodation at any time during the semester. Thus, students entitled to this accommodation are welcome to begin taking learning assessments in the classroom and then notify me if at any point they determine it would be beneficial to move the learning assessments to the testing center. You are required to obtain appropriate documentation and information regarding your accommodations from the Office of Access and Learning Accommodation (OALA). OALA is located in the basement level of the east wing of the Sid Richardson building (reachable via elevator or stairs). They can also be reached by phone (254) 710-3605 or email OALA@baylor.edu.

Classroom Community

Everyone deserves to be treated with respect and dignity in this classroom space: students, teaching assistants, instructors, and guests. Baylor's Code of Ethics (BU-PP 024) states, "Members of the Baylor University community are expected to act in a way that builds a distinctive sense of caring, kindness, mutual respect, collegiality, and fairness. The idea that the Baylor community is a family has always been vital in our historical traditions, and we here reaffirm our commitment to that idea. Out of respect for this community, we do not slander or defame." Incivility, unkindness, and aggression have no place in our classroom community.

Crisis and Emergency Numbers:

Please make a note of the following numbers for crises or emergencies:

- Counseling Center Crisis Line: (254) 710-2467 (Business Hours/Non-Business Hours/Weekends)

- Baylor Police Department: (254) 710-2222
- MHMR Crisis Center: (254) 867-6550
- MHMR 24-Hour Emergency/Crisis Number: (254) 752-3451
- When home during academic breaks, when the counseling center is closed, please call your local resources and/or national hotlines:
 - National Hope Network Hotline: 1-800-SUICIDE (1-800-784-2433)
 - National Suicide Prevention Lifeline: 1-800-273-TALK (1-800-273-8255)

Sexual and Interpersonal Misconduct Policy and Civil Rights Policy

Baylor University does not tolerate unlawful harassment or discrimination on the basis of sex, gender, race, color, disability, national origin, ancestry, age, citizenship, genetic information or the refusal to submit to a genetic test, past, current, or prospective service in the uniformed services, or any other characteristic protected under applicable federal, Texas, or local law (collectively referred to as Protected Characteristics). These policies also prohibit discrimination and harassment based on pregnancy or related conditions.

If you or someone you know would like help related to an experience involving:

1. Sexual assault, sex-based harassment or discrimination, dating violence, domestic violence, stalking, sexual exploitation, or retaliation for reporting one of these types of prohibited conduct, please visit www.baylor.edu/titleix, or contact us at (254) 710-8454, or TitleIX_Coordinator@baylor.edu.
2. Harassment or discrimination (excluding those issues listed in #1) based on Protected Characteristics, please visit www.baylor.edu/civilrights, or contact us at (254) 710-8454 or Civil_Rights@baylor.edu.

For pregnancy or related conditions, the University will provide reasonable modifications to policies, practices, or procedures as necessary to ensure equal access to educational programs and activities. Individuals who need to request reasonable modifications should contact the Office of Access and Learning Accommodation. More information on pregnancy and parenting resources can be found at <https://equity.web.baylor.edu/pregnancy-and-parenting-resources>.

First-Generation Students

Baylor University defines a first-generation college student as a student whose parents did not complete a four-year college degree. The First in Line program is a support office for first-generation college students to utilize if they have any questions or concerns. Please check out First in Line, visit us in Sid Richardson, or email firstinline@baylor.edu.

Military-Connected Students

Veterans, active-duty military personnel, and dependents are encouraged to connect with the VETS program, a space dedicated to supporting our military-connected students. Please communicate, in advance, if possible, any special circumstances (e.g., upcoming deployment, drill requirements, disability accommodations).

Religious Observances

I am committed to providing a caring environment in which students from diverse religious backgrounds can succeed. Please review all the course requirements to identify assignments, exams, or attendance expectations that conflict with religious observance and related cultural practices. I ask that you communicate any conflicts to me within the first week of the semester to allow time for us to discuss the conflict and determine any fair and reasonable adjustments to the schedule and/or assignments. Keep in mind that some changes may simply not be possible for practical reasons of scheduling, but I welcome the opportunity to discuss this with you.

Student Wellbeing

I recognize that students face many struggles during their Baylor career. The Department of CARE Team Services is a group of dedicated and caring case managers who work with students facing mental health issues, financial struggles, and anything else affecting a student's ability to be successful in the classroom. You can find them on the second floor of the Student Life Center, suite 207. Contact Information: (254) 710-2100 ; CareTeam@baylor.edu.

Health Care

Baylor University is strongly committed to addressing the physical well-being and mental health of students by providing access to on-campus healthcare resources. Baylor Health Services includes Primary Care, Psychiatry, Physical Therapy and Pharmacy and is staffed with fully certified and licensed physicians and nurse practitioners, as well as nurses and administrative staff. Appointments may be made by calling their main number or by logging into the health portal located on their website. Contact Information: (254) 710-1010; Health_Services@baylor.edu.

Mental Health

You also have access to mental health services through The Counseling Center. Connect with them for an initial appointment, goal-directed individual therapy, group therapy, nutritional counseling, and case management services. You can schedule your 30-minute initial appointment online through the health portal or by calling (254) 710-2467. If you are experiencing suicidal thoughts, call or text #988 anytime day or night for support.

Telehealth

All Baylor students also have access to medical and counseling services via telehealth in addition to the on-campus services. Medical services include virtual urgent care, psychiatry, and nutrition counseling. Counseling services include 24/7/365 in-the-moment support and ongoing mental health counseling. Access to care is available in the evenings, during weekends, and when the University is closed. All services are free and unlimited to Baylor students. Please visit <https://baylor.academiclivecare.com/> for more information.

Addiction

The Beauchamp Addiction Recovery Center (BARC) supports students in recovery from substance and behavioral addictions through an all-encompassing level of support approach that includes one-on-one mentorship, support groups, and social events open to all Baylor students. Located in the East Village Residential Community (bottom floor of Teal Residential College). Contact Information: (254) 710-7092; BARC@baylor.edu

Food Insecurity

At Baylor, we want all students to have access to food resources that will support their holistic well-being and success. If you or someone you know experiences food insecurity at any time, you can find information on campus and community food resources by visiting The Store. You can also contact Store staff at (254) 710-4931. For additional basic needs assistance, please reach out to CASE or the Care Team.

Spirituality

The Office of Spiritual Life offers programs, persons, and resources to nurture theological depth, spiritual wholeness, and missional living. Located on the corner of 5th and Speight Street in the Bobo Spiritual Life Center. Contact Information: (254) 710-3517; Spiritual_Life@baylor.edu.

How to Use LockDown Browser

Respondus LockDown Browser may be downloaded at:

<http://www.respondus.com/lockdown/download.php?id=755512061>.

There is a “Demonstration Learning Assessment” posted in the Assignment section of Canvas. You can use this assessment at any time to make sure that LockDown Browser is working correctly on your laptop.

LockDown Browser® & Respondus Monitor®

Student Quick Start Guide Canvas Edition

WHAT IS RESPONDUS LOCKDOWN BROWSER?

LockDown Browser is a locked browser for taking quizzes in Canvas. It prevents you from printing, copying, going to another URL, or accessing other applications during an assessment. If a Canvas quiz requires that LockDown Browser be used, you will not be able to take the quiz with a standard web browser.

LockDown Browser should only be used for taking Canvas Quizzes. It should not be used in other areas of Canvas.

INSTALLING LOCKDOWN BROWSER

If LockDown Browser has already been installed, skip to the next section. If not, LockDown Browser must be installed to each computer (Windows or Mac) being used to take a test.

- Your institution or instructor will provide the link for downloading and installing LockDown Browser.
- Follow the onscreen instructions to complete the install.

TAKING AN ASSESSMENT

1. Close all programs, unless one is used to connect you to the Internet.
2. Locate the “LockDown Browser” shortcut on the desktop and double-click it. (For Mac users, launch “LockDown Browser” from the Applications folder.)
3. If prompted to close a blocked program (e.g. screen capture, instant messaging) choose **Yes** when prompted.
4. Login to your Canvas course.
5. Navigate to the quiz and select the **Take a Quiz** link.
6. Once a quiz has been started with LockDown Browser, you cannot exit until the **Submit Answers** button is clicked.

USING WITH A WEBCAM (RESPONDUS MONITOR)

You may be required to use LockDown Browser with a webcam, which will record you during an online, non-proctored exam. (The webcam feature is sometimes referred to as “Respondus Monitor.”)

Your computer must have a functioning webcam and microphone. A broadband connection is also required.

If an exam requires LockDown Browser **and** a webcam, follow steps 1-5 in the previous “Taking A Test” section. At this point the Startup Sequence for the webcam begins.

- You will first need to review and agree to the Terms of Use.
- The **Webcam Check** will confirm that your webcam and microphone are working properly. The first time the Webcam Check is performed on a computer, Adobe Flash Player will require you to select **Allow** and **Remember**.
- The remaining steps of the Startup Sequence will depend on settings chosen by your instructor. Follow the instructions and note your progress along the top of the screen.
- If you encounter a problem, select the **It's not working** link for troubleshooting tips.

The test will begin after the Startup Sequence is complete. You cannot exit LockDown Browser until the test is submitted for grading.

PROBLEMS?

If you have problems downloading, installing, or taking a test with Respondus LockDown Browser, contact your instructor or your institution's help desk.

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Note that, although the LockDown Browser is most often used in conjunction with a *webcam* (Respondus Monitor) for completing tests *outside* of class, we will be using the program **without** a webcam for completing learning assessments **during** class. For additional details on using LockDown Browser, go to:

<http://www.respondus.com/products/lockdown-browser/guides.shtml>.

PSY 4400 Course Calendar, Fall 2025

	Aug 26 First day of class.		Aug 28 1 Start 30 min Learning Assessments. Add Study Guide 0A-1A questions
Sep 1 Begin practice quizzes	Sep 2 2 Add Study Guide 1B questions		Sep 4 3 Add Study Guide 1C questions
Sep 8 Homework 1 due	Sep 9 4 Add Study Guide 2A questions Add Homework 1 questions <i>Project 1 lab session</i>	Sep 10 <i>Suggested: try Project 1</i>	Sep 11 5 Add Study Guide 2B questions
Sep 15 Project 1 due	Sep 16 6 Add Study Guide 2C questions (A)		Sep 18 7 Add Study Guide 2D questions Add Unit 1 written summaries
	Sep 23 8 Add Study Guide 2E questions		Sep 25 9 Add Study Guide 2F questions
Sep 29 Homework 2 due	Sep 30 10 Start 35 min Learning Assessments. Add Study Guide 3A questions Add Homework 2 questions <i>Project 2 lab session</i>	Oct 1 Project 2 due	Oct 2 11 Add Study Guide 3B questions
	Oct 7 12 Add Study Guide 3C questions Add Unit 2 written summaries (B)		Oct 9 13 Add Study Guide 3D questions
Oct 13 Homework 3 due	Oct 14 14 Add Study Guide 4A questions Add Homework 3 questions	Oct 15 Project 3 due	Oct 16 15 Add Study Guide 4B questions
	Oct 21 16 Add Study Guide 4C questions Add Unit 3 written summaries (C)		Oct 23 17 Add Study Guide 4D questions
Oct 27 Homework 4 due	Oct 28 18 Add Study Guide 5A questions Add Homework 4 questions (D)		Oct 30 19 Add Study Guide 5B questions
	Nov 4 20 Add Study Guide 5C questions (E)		Nov 6 21 Add Study Guide 5D questions
Nov 10 Homework 5 due	Nov 11 22 Add Study Guide 6A questions Add Homework 5 questions (F)	Nov 12 Project 4 due	Nov 13 23 Add Study Guide 6B questions
	Nov 18 24 Add Study Guide 6C questions Add Unit 5 written summaries (G)		Nov 20 25 Add Study Guide 6D questions
	Nov 25 (Thanksgiving break)		Nov 27 (Thanksgiving break)
Dec 1 Homework 6 due	Dec 2 26 Add Homework 6 questions (H)		Dec 4 27 (Nothing new added.)
	Dec 9 Make up Learning Assessments		

The final exam will be on Wednesday, December 17 at 4:30 PM.